# Experiment 10: Study of Malicious Software Using Different Tools

## Aim

To study malicious software techniques and perform security testing using different tools:

a) Keylogger attack demonstration

b) DoS attack simulation

c) Vulnerability scanning using Nessus or Kali Linux tools

## Tools Required

Operating System: Microsoft Windows

Virtualization Software: VirtualBox / VMware (recommended)

Kali Linux (optional but useful for DoS simulation)

Keylogger Tool: Revealer Keylogger / Spyrix

DoS Tool: Hping3 / Nmap

Vulnerability Scanner: Nessus

Internet Connection

## Theory

Malicious software (Malware) is software designed to harm a system, steal information, or gain unauthorized access.

Studying malware techniques helps students understand cybersecurity threats and protection methods.

In this experiment we study:

1. Keylogging – capturing user keystrokes

2. DoS attack – flooding a system with network packets

3. Vulnerability scanning – identifying weaknesses in a system

These experiments must be performed only in a controlled lab environment.

## Part A: Keylogger Attack (Windows)

Step 1: Download a Keylogger Tool

Download an educational keylogger such as Revealer Keylogger or Spyrix Free Keylogger from its official website.

Step 2: Install the Software

Run the setup file and complete installation.

Step 3: Launch the Keylogger

Open the installed keylogger application and enable logging.

Step 4: Generate Activity

Open Notepad or a web browser and type some text such as:

Username: student123

Password: abc123

Step 5: View Recorded Data

Open the keylogger dashboard and observe:

• Recorded keystrokes

• Time of typing

• Application used

## Part B: DoS Attack Simulation

Step 1: Identify Target IP Address

Open Command Prompt and run:

ipconfig

Note the IP address of the target machine.

Step 2: Check Connectivity

Run:

ping target_ip

Step 3: Run DoS Simulation

If using Kali Linux or WSL:

hping3 -S --flood -p 80 target_ip

Explanation:

-S : Sends SYN packets

--flood : Sends packets continuously

-p 80 : Target port

Step 4: Observe Results

The target system may show increased network traffic or slower response.

## Part C: Vulnerability Scanning Using Nessus

Step 1: Install Nessus

Download Nessus from the official website and install it.

Step 2: Open Nessus

Open browser and go to:

https://localhost:8834

Step 3: Login and Setup

Create account and login to Nessus dashboard.

Step 4: Create New Scan

Click:

New Scan → Basic Network Scan

Enter:

Scan Name

Target IP Address

Step 5: Start Scan

Click Launch to begin scanning.

Step 6: Analyze Results

Nessus will display:

• Critical vulnerabilities

• High risk issues

• Medium risk issues

• Open ports and services

## Observation Table

| Activity | Tool Used | Purpose |
|---|---|---|
| Keylogger Attack | Keylogger Tool | Record user keystrokes |
| DoS Simulation | Hping3 / Nmap | Generate heavy network traffic |

| | | |
|---|---|---|
| Vulnerability Scan | Nessus | Detect system vulnerabilities |

## Result

The experiment was successfully performed. The keylogger recorded keystrokes, DoS attack simulation generated heavy traffic, and vulnerability scanning detected system weaknesses.

## Conclusion

This experiment helped students understand different types of cyber attacks and security analysis tools used in real-world cybersecurity testing.